

SpotDrop

A Spotify Feature Case Study

Phase 2a: Product Requirements Document

Mayank Pandey

M.S. Engineering Management, University of Southern California

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Product: Spotify

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1. Overview

SpotDrop is a proposed Spotify feature that makes music discovery social and ambient. By enabling users to see what people nearby are listening to in real time, using proximity technology similar to Bluetooth and AirDrop, SpotDrop creates a curiosity-driven discovery experience that connects people through music without requiring a verbal introduction.

Users choose to remain anonymous or visible when sharing their listening activity. A commuter on a train can send a request to listen together with someone nearby, browse what others are playing without revealing their identity, or save a track from a stranger's queue directly to their library. Spotify can also surface SpotDrop activity through recap features such as Your Sound Capsule, adding a social dimension to existing engagement metrics.

Business Goals

- Increase user engagement and time spent in the app, particularly in shared public spaces.
- Strengthen Spotify's position as a leader in social and community-driven music experiences.
- Drive music discovery and playlist creation through peer-to-peer channels.

Design Principles

- Privacy first: every interaction is opt-in and controllable at all times.
- Frictionless: core actions must be achievable in 2 taps or fewer.
- Non-intrusive: SpotDrop should enhance the listening experience, never interrupt it.

2. Goals and Success Metrics

The following KPIs are organized by the three primary product goals. All targets are set for a controlled pilot launch in 1 to 2 high-density markets prior to global rollout.

Goal Area	Metric	Target	Timeframe
Engagement	Feature Adoption Rate	15% of DAU try SpotDrop	First 3 months
Engagement	Jam Request Send Rate	25% of SpotDrop users send at least 1 request	First 3 months
Engagement	Jam Request Acceptance Rate	40% of requests accepted	First 3 months
Engagement	DAU/MAU Ratio	Greater than 30% for SpotDrop users	Ongoing
Engagement	Average Session Length	2 to 3 minutes added on top of core listening time	First 3 months
Engagement	Weekly Retention Rate	35% of first-time users return within 90 days	First 90 days

Engagement	Monthly Churn Rate	Below 20% after first month	Month 2 onward
Engagement	NPS Score	7/10 or above in post-launch survey	60 days post-launch
Communication	Jam Request Volume	500k daily requests globally	6 months post-launch
Communication	Active Jam Sessions	200k concurrent daily sessions globally	6 months post-launch
Communication	Request Response Latency	90% of requests responded to within 10 seconds	6 months post-launch
Discovery	Songs Saved via SpotDrop	At least 1 new song saved per 5 SpotDrop users per week	Ongoing
Discovery	Artist Diversity Rate	20% of saved songs from artists outside the top 100	Ongoing
Discovery	Library Growth Lift	5 to 8% more songs saved per month vs. non-SpotDrop users	Month 3 onward
Discovery	Sound Capsule Integration	3 new SpotDrop stats added to Your Sound Capsule	Launch

3. User Personas

Full persona documentation is provided in Phase 1: Discovery and Research. A brief summary of the two primary personas is included here for reference.

Rahul, The Curious Commuter

Age 23, university student. Commutes daily by metro. Feels isolated in crowded public spaces and has no natural way to discover music through the people around him. His primary need is a low-friction, anonymous way to see what others are listening to and occasionally connect over shared taste.

Sarah, The Social Explorer

Age 26, freelance designer. Works from cafes and co-working spaces. An active music curator who finds algorithmic recommendations too predictable. Her primary need is a way to receive and broadcast music taste to like-minded people in her physical environment.

4. User Stories and Acceptance Criteria

Each user story follows the format: As a [persona], I want to [action] so that [outcome]. Acceptance criteria define the minimum conditions for a story to be considered complete.

Feature	User Story	Acceptance Criteria	Priority
Passive Discovery	As a commuter, I want to see what music people nearby are listening to so that I can satisfy my curiosity and discover new songs organically.	Nearby feed displays real-time listening activity with song, artist, and distance. Updates every 30 seconds.	Must-Have
Privacy Control	As a user, I want to toggle SpotDrop on and off instantly so that I stay in full control of when my listening is visible to others.	Active/Inactive toggle accessible from the main SpotDrop tab in one tap. Status persists until changed.	Must-Have
Anonymous Mode	As a privacy-conscious user, I want to browse nearby listeners anonymously so that I can discover music without revealing my identity.	Anonymous chip on main feed. In anonymous mode, username and avatar are hidden; song is still visible to others if active.	Must-Have
Save to Library	As a user, I want to save a song I see someone nearby playing directly to my Spotify library so that I can listen to it again without losing track of it.	Save to Library button on user profile card and in jam session. Confirms with a toast notification.	Must-Have
Jam Request	As a curious listener, I want to send a non-verbal listen-together request to someone nearby so that we can share the same music in real time without a verbal introduction.	Listen Together button on user profile triggers a non-blocking pop-up for the recipient. Recipient can Accept or Decline.	Must-Have
Session Management	As a user, I want to decline jam requests and end a session at any time so that I am never forced into a shared listening experience I did not choose.	Decline button on incoming request. End Jam button always visible during active session. No penalty for declining.	Must-Have
Ephemeral Chat	As a user in a jam session, I want to send short messages and emoji reactions so that I can express how I feel about a song without exchanging contact details.	Lightweight text bar in jam session. Messages limited to short phrases. Chat clears automatically when session ends. Reactions are anonymous.	Should-Have
Playlist Sharing	As a user, I want to share my top artists and songs via a lightweight profile so that others nearby can get a quick sense of my music taste.	Profile card displays top 5 artists and top 5 songs. Visible when user taps a listener card from the feed.	Should-Have
AI-Powered Save	As a user ending a jam session, I want to save the session as a named playlist so that the experience becomes a lasting memory I can revisit.	End Jam modal offers Create Playlist option. AI generates a creative playlist name based on the songs played. One-tap save to Spotify library.	Could-Have

Alternate Play	As a user in a jam session, I want to take turns adding songs to the queue so that both participants shape the listening experience equally.	Add Song button available to all jam participants. Queue shows who added each track.	Could-Have
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5. Feature Prioritization (RICE Framework)

Features are scored using the RICE framework: (Reach x Impact x Confidence) divided by Effort. Reach is scored on a 1 to 10 scale representing estimated monthly users. Impact uses the standard scale: 3 = massive, 2 = high, 1 = medium, 0.5 = low. Confidence is expressed as a percentage reflecting certainty in the estimates. Effort is measured in person-months.

Feature	Reach	Impact	Confidence	Effort	RICE Score	Priority	Rank
Nearby Listening Feed	10	3	80%	2	12.0	Must-Have	1
Privacy Toggle	10	3	100%	1	30.0	Must-Have	2
Save/Like to Library	9	2	100%	1	18.0	Must-Have	3
Jam Request (Send/Accept)	8	2	80%	3	4.3	Must-Have	4
Anonymous Mode	8	2	100%	2	8.0	Must-Have	5
Lightweight Profiles	7	1	80%	2	2.8	Should-Have	6
Emoji Reactions	6	1	80%	1	4.8	Should-Have	7
Ephemeral Chat	5	1	80%	2	2.0	Should-Have	8
AI Icebreakers	5	2	60%	3	2.0	Could-Have	9
Sound Capsule Integration	7	1	80%	2	2.8	Could-Have	10
Silent Disco Mode	4	2	60%	4	1.2	Could-Have	11
Artist Spotlight	3	1	50%	3	0.5	Won't-Have (v1)	12
Group Jams (3+ users)	5	2	60%	5	1.2	Won't-Have (v1)	13

Features scoring above 8.0 are prioritized for the MVP. Features scoring 2.0 to 8.0 are scheduled for the Next phase of the roadmap. Features below 2.0 are deferred to the Later phase or flagged for further research before commitment.

6. Assumptions and Constraints

Assumptions

- Users in the target demographic (18 to 34) are comfortable with opt-in, anonymous location-adjacent sharing in public contexts.
- Spotify's existing infrastructure can support real-time proximity data and music session syncing at scale with targeted engineering investment.
- Initial adoption will be driven by a clear value proposition, low-friction onboarding, and integration with existing features such as Your Sound Capsule.
- Peer-driven discovery resonates most strongly in high-density urban environments (commuter transit, university campuses, cafes).

Constraints

- SpotDrop must not compromise user privacy or security under any conditions. Privacy controls are non-negotiable and must be included in the MVP.
- The feature must not degrade the core listening experience. SpotDrop interactions must run independently of the main playback engine.
- Jam sessions must maintain sub-200ms synchronization latency to deliver a seamless shared listening experience.
- The feature must be compliant with GDPR, CCPA, and applicable regional data regulations from day one.

7. Technical Requirements

Requirement	Specification
Spotify Integration	SpotDrop appears as a dedicated tab in the bottom nav bar, to the right of Your Library. Interacts directly with Spotify playback, sharing, and user profile APIs. Supports seamless transition between core listening and SpotDrop sessions.
Proximity Detection (iOS)	Uses Apple AirDrop framework for peer-to-peer device discovery and direct sharing without an internet connection.
Proximity Detection (Android)	Uses Bluetooth Low Energy (BLE) and Wi-Fi Direct for detecting nearby SpotDrop users.
Offline Mode	SpotDrop operates in low-connectivity environments (underground subways, tunnels). Jam requests and song discovery sync locally via peer-to-peer protocols. Server sync occurs once the device reconnects to the network.
Real-Time Communication	WebSockets used when internet is available, targeting sub-100ms latency for jam synchronization. Fallback to AirDrop, BLE, or Wi-Fi Direct in offline environments.
Notifications	Non-intrusive pop-up notifications for jam requests, queue updates, and song additions. Notifications disappear automatically after 10 seconds if

ignored. Must not interrupt active audio playback.

8. Non-Functional Requirements

Category	Requirement
Performance	Real-time updates delivered in under 200ms when a user adds a song to the jam queue. Pop-up notifications display song title, username (if not anonymous), and countdown to next track.
Scalability	Supports small sessions (2 to 5 users) and large-scale environments (silent discos, festivals). Architecture must sustain up to 10,000 concurrent jam sessions globally without service degradation.
Reliability	Target uptime of 99.9% across all devices. Offline caching allows continued viewing of the nearby feed and jam request exchange without network access.
Security	All proximity connections encrypted with AES-256 (Wi-Fi/BLE) and Apple secure AirDrop protocol (iOS). No chat logs or personal data stored on Spotify servers. Session chat is ephemeral and deleted automatically when the jam ends. GDPR and CCPA compliant.
Usability	Core actions (start a jam, save a song, send a request) achievable in 2 taps or fewer. WCAG 2.1 AA accessibility compliance. Notifications are contextual and non-intrusive.
Identity Control	Three sharing modes: Anonymous (music only), Partial (music and top artists), Full (music, profile, and bio). User can switch modes at any time from the main feed.

9. Future Enhancements

The following capabilities are out of scope for v1 but are documented here to inform architecture decisions and ensure the system is designed to accommodate them.

Group Jams

Allow three or more users to create a shared queue collaboratively in real time, with a voting mechanism to determine the next track.

Extended Profiles

Beyond the lightweight v1 profile, users could share mood tags, recent listening history, and profile badges for frequent jammers and music trendsetters.

Artist Spotlight Integration

When songs from non-top-100 artists are frequently discovered via SpotDrop, Spotify surfaces those artists in a dedicated spotlight, connecting the feature to Spotify's emerging artist mission.

Event-Based SpotDrop

Integration with concerts, festivals, and club venues. Verified venue profiles host official playlists; attendees can join and sync in real time, enabling large-scale silent disco experiences.

AI-Powered Recommendations

Based on SpotDrop interactions, Spotify generates personalized recommendations that incorporate local discovery signals alongside algorithmic behavior signals.

CarPlay and Android Auto Integration

Safety-focused SpotDrop for in-vehicle use. Drivers and passengers in nearby cars can discover what others are listening to and send or receive jam requests using simplified UI and voice commands.

Cross-Platform Expansion

SpotDrop functionality extended to the Spotify Web Player, desktop application, and smart speakers for users who primarily listen outside mobile.

10. Risks and Mitigations

Risk	Level	Description	Mitigation
Privacy Concerns	High	Users hesitate to share location or listening activity.	Opt-in only design, strong anonymity options, clear consent flows, and granular privacy settings.
Battery and Performance	Medium	Continuous BLE, Wi-Fi, or location use drains battery or degrades device performance.	Optimize background processes. Offer Always On and While Using modes so users choose their trade-off.
Low Adoption	Medium	Users do not engage with SpotDrop or treat it as a passive curiosity rather than a core habit.	In-app onboarding tutorials, Sound Capsule integration, and gamification to build habitual use.
Network Limitations	Medium	Real-time features degrade or fail in low-connectivity environments.	BLE fallback for offline proximity detection. Server sync on reconnect.
Spam and Misuse	Medium	Excessive or unwanted jam requests create a poor experience.	Spam filters, block and report functionality, and per-session usage limits.
Regulatory Compliance	High	Regional regulations may restrict proximity sharing, data usage, or music licensing in certain markets.	Legal review per market prior to launch. GDPR and CCPA compliance built into architecture from day one.
Apple Ecosystem Friction	Low	Apple may challenge SpotDrop's AirDrop-like	Distinct UI patterns, clear positioning as a music feature rather than a file-transfer

		interface during App Store review.	tool, and proactive Apple developer relations engagement.
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11. Closing Statement and Next Steps

SpotDrop represents a natural and defensible evolution for Spotify. It does not require users to leave the platform, does not compete with Spotify's existing algorithmic engine, and addresses a real problem that no current Spotify feature solves: the isolation of listening in shared physical spaces.

The combination of proximity-based discovery, opt-in social connection, and deep integration with existing Spotify features (library, Sound Capsule, playback API) positions SpotDrop as a high-leverage feature that increases engagement, drives discovery, and builds emotional connection to the platform.

Immediate Next Steps

1. Complete the Phase 1 user research survey to validate persona assumptions and adoption intent targets.
2. Conduct technical feasibility review for BLE and AirDrop proximity detection with Spotify platform engineering.
3. Finalize Phase 2b product roadmap and design mockups for core user flows.
4. Engage legal and privacy teams for early GDPR and CCPA compliance review.
5. Identify pilot market candidates: target 1 to 2 high-density urban markets with strong transit culture for controlled launch.